Phase-field simulation is a computer method for studying how materials change over time. It can show how particles grow, how boundaries move and how tiny pores form or disappear. This makes it useful for nanomaterials research, including nanoparticles, thin coatings and materials with nanoscale features.

One application is solid-state batteries. These batteries depend on good contact between lithium metal and the solid electrolyte. Very small gaps, including nanovoids, can interrupt that contact and make it harder for current to pass. A published phase-field study has modeled nanovoids and their growth at these interfaces.

Our paper studies a related contact problem using a model of pressure and electrical resistance. It examines how pressure changes and the contact present at the start of discharge affect resistance. Its link to nano applications is the role of small gaps and interface design, where nanoscale coatings or interlayers can help improve contact.

More generally, phase-field simulations can compare material structures, coating properties and pore sizes before experiments. For a nanomaterials program, this helps connect the design of small features to how a device performs and guides which materials are worth making and testing.